

Based on the current Soneium architecture, the existing flow is **Media** → **Fingerprint** → **CopySight Analysis** → **Analysis Fingerprint** → **Rule Enforcement** → **Soneium**, with a minimal four-field immutable attestation and invalid submissions reverting before anything is written on-chain.

PRD — Wallet Ownership & Royalty Settlement Upgrade

Objective

Extend the existing CopySight → Soneium flow with:

1. **Wallet-linked asset ownership**
2. **On-chain rights-holder attribution**
3. **Automatic royalty calculation based on CopyScore**
4. **Mandatory royalty payment before blockchain registration**

The blockchain record must **not be created unless the required royalty transaction succeeds**.

Updated Flow



└ YES



Soneium Attestation

For a **clean asset with no detected match / CopyScore = 0**, the asset fingerprint is linked to the connected wallet as its rights-holder claim.

Wallet control alone should not be described as legal proof of copyright ownership. For MVP, it represents the **on-chain rights-holder/ownership claim** attached to the asset.

Functional Requirements

1. Wallet Connection

Before blockchain submission:

- User connects an EVM-compatible wallet.
- CopySight stores the wallet against the user's profile/session.
- User signs a message proving control of the wallet.
- Asset fingerprint is associated with the submitting wallet.

Required data:

user/profile ID
wallet address
asset fingerprint
wallet signature
timestamp

2. Rights Registry

Add a **Rights Registry** between fingerprinting and final submission.

For every fingerprint, determine whether:

A. No existing rights holder exists

Asset is clear.

assetFingerprint → connectedWallet

Create an ownership/right-holder claim for that wallet.

B. Existing rights holder exists

Return:

rightsHolderWallet
assetFingerprint
license/IP price
payment token
rights record ID

This should be additive rather than replacing the existing CopySight analysis record, which is currently deliberately minimal and immutable.

3. Royalty Calculation

When CopySight detects protected IP:

Royalty = IP Rights Price × CopyScore / 100

Example:

CopyScore: 75
IP Rights Price: \$100

Royalty: \$75

The UI displays before submission:

CopyScore: 75%
Rights holder: 0x123...
Base IP price: \$100
Required royalty: \$75

[Pay & Register]

For MVP, the rights price must come from the **Rights Registry**, rather than being entered by the submitting user.

4. Payment + Blockchain Submission

Preferred architecture: make **payment and registration atomic**.

User Wallet



Settlement Contract

- └─ transfer royalty → Rights Holder
- └─ create/authorize CopySight attestation

The user receives a transaction request directly in their wallet.

If:

wallet rejected
insufficient balance
insufficient token approval
payment reverted
wrong amount

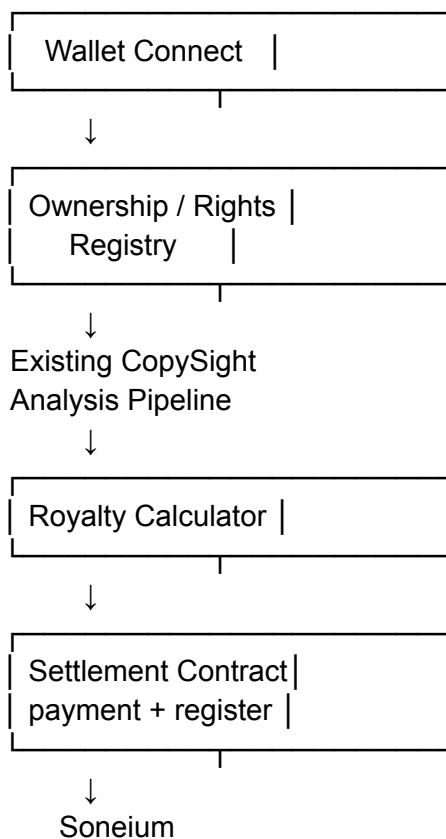
then:

Royalty = unpaid
Attestation = NOT created
Asset = NOT registered on-chain

Only a confirmed successful payment allows the existing attestation flow to complete.

This preserves the current principle that failed rule checks result in nothing being written to Soneium.

New Architectural Modules



Contracts / components

Rights Registry

- `assetFingerprint`
- `rightsHolder`
- `baseRightsPrice`
- `paymentToken`
- `rightsVersion`

Settlement Contract

- Calculates/verifies royalty
- Transfers funds
- Prevents registration without payment
- Emits payment/settlement event
- Triggers or authorizes the existing attestation flow

Existing Rule Resolver

- Keep current signer validation
 - Keep CopyScore validation
 - Keep data-shape validation
 - Add validation that required settlement has completed
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MVP Frontend

Frontend can be vibecoded using **copyright.ai as the visual reference**.

Only new states are required:

Connect Wallet

↓

Asset Owner
0x...1234

↓

CopySight Result
CopyScore: 75%

↓

IP Rights Detected
Rights Holder: 0x...5678
Rights Price: \$100
Royalty Due: \$75

[Pay \$75 & Register]

↓

Wallet Transaction

↓

✓ Payment Confirmed
✓ Asset Registered on Soneium

Failure:

Transaction declined

Asset was NOT registered on-chain.

[Try Again]

MVP Acceptance Criteria

- User cannot enter the blockchain registration flow without connecting a wallet.
- Every new asset is associated with the submitting wallet.
- Clean assets can generate a rights-holder claim linked to that wallet.
- Existing rights-holder assets resolve the correct holder and licensing price.
- Royalty is calculated automatically from **CopyScore × Rights Price**.
- User receives the transaction request in their connected wallet.
- Required royalty is transferred directly to the rights holder.
- Failed/declined payment creates **no blockchain registration**.
- Successful payment results in the existing immutable CopySight/Soneium record.
- Payment, rights holder and resulting attestation can be deterministically linked for later verification.

The existing system already tracks submitted/confirmed/failed states and protects against duplicate submissions, so that lifecycle should be reused rather than rebuilt.